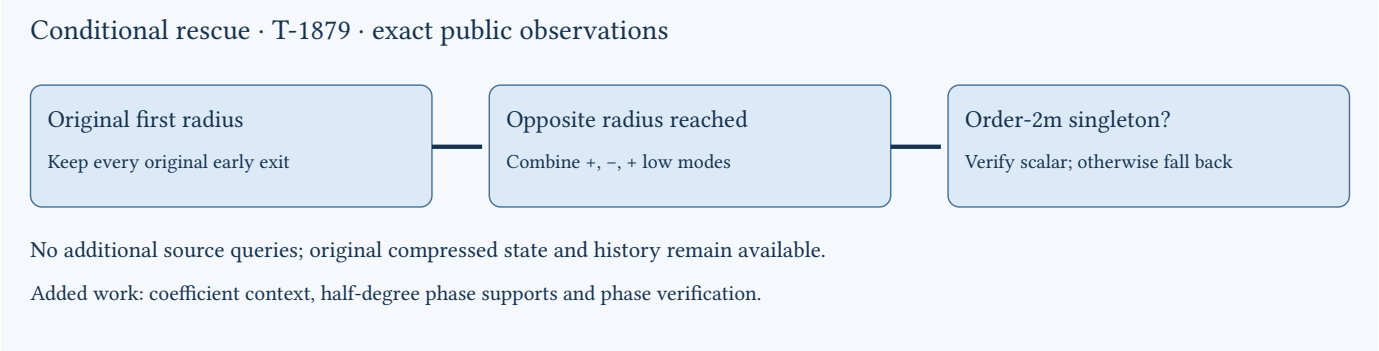
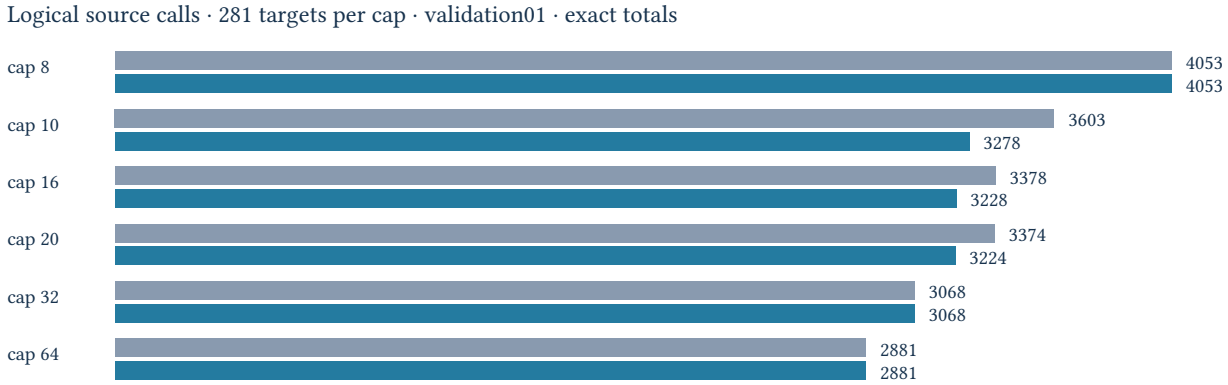
Conditional doubled-order phase rescue

T-1879 · Prime-field compressed recovery · 2026-11-07

For odd m and 8m dividing r−1, opposite m-point orbits form one 2m-point orbit. Their three low observations combine with signs +, −, +. The new helper uses that identity after the first-radius path defers, and keeps the original state as fallback.



Identity checks. 10,503 scalar-mode values; 5,103 reindexed sign vectors (exhaustive for m=3, partial for m=5); 21 coefficient sections; 4,050 wrong-sign controls; 10 invalid inputs. Fifteen nonopposite public fallbacks match the original path.



Public panel. E/F_{257}: $y^2=x^3+x+7$; P=(1,3); r=281; m=5. Radii 1,280,2,3; extras 0 through 16; call cap 128. Every target/cap pair has no source-call increase and preserves every reference success. Five caps recover all 281 targets; cap 8 leaves 10 unresolved on both paths. At cap 64 no rescue context is constructed.

Added work. Caps 10,16,20 build 260 extra contexts and deliver 2,080 upper F0 coefficients each. Cap 32 builds 205 contexts, with 1,640 such deliveries. These exclude F1/F2 and setup. Cap misses and nonsingletons are retained. Exact-order checks, polynomial work, copies, allocations and final group verification are charged; query savings are not a runtime claim.

Evidence. validation01 passes on its first attempt with 94 native dependency hashes and inherited checks. PCC-20261107-67816b is a producer attestation; independent signed verdict pending. No cloud or timed workload. Original cohort unchanged. DERIVATION.md, RESULTS.md and README.md provide scope and replay.

Sources and next. Exact paired/direct predecessor; NIST DLMF 26.3; FFTW transform-definition conventions. Finite-field recombination is derived explicitly; KB unavailable. Next reduce or reuse the added coefficient setup, then account total arithmetic before default adoption.